

# Pedro Madruga

Principal AI Engineer · Building AI products and teams · From idea to production and scale, twice

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Principal AI Engineer and AI leader. Twice, I have taken an AI product from idea through production and scaling, building and leading the team behind each. KAILA serves 5k+ daily users across four countries; KAILA FLOW launched in all four in June 2026, with paying customers in each during launch week. FLOW is Karnov Group's largest AI project and KAILA its second largest; together, they account for a major share of the Group's total revenue. My approach is Domain Driven AI: embedding domain experts into development so the technology serves real workflows.

## PROFESSIONAL EXPERIENCE

### **Principal AI Engineer, Group** - Karnov Group AB

Oct 2024 - Present · Copenhagen

- Co-created KAILA FLOW and led it from concept through Karnov's largest product validation to production in four countries in June 2026, with paying customers in each during launch week
- Built and lead the FLOW team, with six current direct reports and people and budget responsibility. Lead multi-agent workflows for case intake, planning, source-grounded research and document drafting
- Continued as AI lead for KAILA / JunoAI, scaling from 3.5M to ~25M documents across four markets using RAG and Agentic DeepSearch. My retrieval engine remains its foundational core; used by leading law firms and the highest courts in Denmark and Sweden
- Drive Group-level AI research and strategy with domain experts, Architects, Software Engineers and DevOps

### **Senior AI Engineer** - Karnov Group AB

Oct 2022 - Sept 2024 · Copenhagen

- Created KAILA as Karnov's only AI Engineer and led it from concept to production. Delivered the legal research assistant in six months and scaled its corpus from 5k to 3.5M documents
- Designed and built the retrieval engine that remains KAILA's foundational core, working with domain experts on legal accuracy
- Recruited every member and grew the AI team from one to six; architected the embeddings pipeline, cloud infrastructure and AI engineering practices
- Analyzed millions of search queries and developed proofs of concept for business stakeholders

### **Senior Data Scientist** - GoSimplo

Nov 2021 - Sep 2022 · Copenhagen

- Built deep learning forecasting systems (Keras/TensorFlow) for financial forecasting used by hundreds of users
- Developed time-series forecasting systems (XGBoost, ARIMA) deployed via Docker on Azure
- Processed large-scale data with Azure Synapse/SQL, delivered Streamlit dashboards

### **Data Scientist & Founder** - JohnConnor.ai ApS

Jun 2020 - Nov 2021 · Copenhagen

- Founded an AI/ML consultancy serving GrantCompass and Karnov Group
- Built recommendation systems with AWS Personalize and data warehouses integrating multiple sources

## EARLIER EXPERIENCE

### **Data Scientist** - Karnov Group Denmark

May 2018 - May 2020 · Copenhagen

- Developed content-based recommendations for Danish legal documents using spaCy and Prodigy
- Identified browser performance bottlenecks through data analysis, helping the engineering team reduce response times from 30 seconds to 2 seconds

### **Senior Software Engineer** - Karnov Group Denmark

Apr 2017 - Apr 2018 · Copenhagen

- Developed full-stack applications (React/Node.js), deployed on GCP/Kubernetes; built Karnov Business Optimizer

**Earlier software engineering and data analysis experience (2006-2017):** Founder & Software Engineer at Occam.ooo (2016-17); Software Engineer at colego.dk (2015-16, including team leadership) and Unwire (2014-15, Nordea payment apps); Data Analyst at Copenhagenize Design Co. (2012-13, research featured in Business Insider); part-time Web Developer at Produtrónica (2006-12) during studies.

## EDUCATION

**MSc + BSc Environmental Engineering**
Universidade Nova de Lisboa
2006 - 2012

**Coursework in Computer & Software Engineering**
Instituto Superior de Engenharia de Lisboa
2003 - 2005

## TECHNICAL SKILLS

**AI:** Multi-agent systems, RAG, Information Retrieval, LangGraph, Pydantic AI, Langfuse
**Engineering:** Python, Node.js, gRPC, Claude Code (spec-driven)
**Cloud & Deployment:** GCP, Azure, Docker, Kubernetes
**Course:** DTU 02450 Machine Learning (40 hours, 2019)

## LANGUAGES

**Danish:** Conversational · **English:** Native/Bilingual · **Portuguese:** Native · **Spanish:** Professional · **French:** Limited Working

## LEADERSHIP & RECOGNITION

- Established Karnov's initial AI engineering capability and personally built and led both product teams. The wider AI organization has grown from me alone in 2022 to ~20 Engineers and Architects
- Established internal AI knowledge-sharing through talks on new technologies and engineering practices
- KAILA covered in Danish tech and professional press, including Version2/Ingeniøren, Finansforbundet and Djøfbladet

## ADDITIONAL INFORMATION

**Open Source:** Creator of react-native-clean-project (1.3K+ GitHub stars); contributor to multiple Open Source projects. **Certification:** SQL for Data Science (Coursera, 2020). **Interests:** Family, long-distance running (inc. ultramarathons), reading non-fiction, hacking personal projects.